

Team-discussion prompts: Traditional ML methods, Embeddings, and Evaluation

Topic A: More word embeddings

Task: What are techniques beyond Word2Vec and GloVe?

1. Read up on FastText and ELMo. What are the differences? Generate a table to compare and contrast the characteristics of these approaches?
2. Show some code examples to apply these different types of word embedding methods.
3. Which is better? Any specific context(s)? Why? How?

Resources:

- <https://www.alpha-quantum.com/blog/word-embeddings/introduction-to-word-embeddings-word2vec-glove-fasttext-and-elmo/>
- <https://medium.com/@kashyapkathrani/all-about-embeddings-829c8ff0bf5b>
- <https://medium.com/nlplanet/two-minutes-nlp-11-word-embeddings-models-you-should-know-a0581763b9a9>

Topic B: Different classifiers

Task: Choose a different machine learning classifier NOT discussed in the lecture

1. For example, Decision Tree, Logistic Regression, XGBoost, Random Forest, Ensemble (e.g., Bagging, Boosting, Voting)
2. Try out Python codes for any labeled text data such as 20NewsGroup (supervised approach) to demonstrate your understanding of the working principles.
3. To show the performance, which metric is the best: precision, recall, AUC-ROC, F1 (or F2?) and in which situation(s)?

Resources:

- <https://www.kaggle.com/code/heeraldedhia/text-classification-nlp>
- <https://www.kaggle.com/code/diveki/classification-with-nlp-xgboost-and-pipelines>
- <https://ai.plainenglish.io/sentiment-classification-using-xgboost-7abdaf4771f9>
- <https://www.analyticsvidhya.com/blog/2020/09/precision-recall-machine-learning/>
- <https://aclanthology.org/2020.eval4nlp-1.9/> (optional further reading)

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Topic C: Clustering visualisation

Task: Choose either hierarchical or fuzzy clustering methods

1. Use any text dataset and generate Python codes to show the step-by-step process of how clusters are being formed (e.g., agglomerative versus divisive)
2. Print out the dendrogram for better visualisation
3. What are the pros and cons (advantages and limitations) of hierarchical methods as compared to k-means?

Resources:

- https://colab.research.google.com/github/dipanjanS/nlp_workshop_odsc19/blob/master/Module05%20-%20NLP%20Applications/Project02%20-%20Text_Clustering.ipynb
- <https://medium.com/@fahdbalogun.a/agglomerative-clustering-for-text-1d9d2b17124f>
- <https://medium.com/@mohamad.razzi.my/hard-clustering-vs-soft-clustering-in-nlp-da8f1f2ba2ec>

Topic D: Compare metrics

Task: Try out either summarization or translation

1. Construct predicted and target/reference (ground truth) sentences
2. Evaluate Rouge-N, Rouge-L, and Rouge-S scores. Which of the 3 metrics works best for your example(s)? When are they different? Show examples.
3. Compute both BLEU and METEOR scores. Discuss how they differ, and which is preferred if any. Try to find examples where BLEU is preferred over METEOR and vice versa

Resources:

- <https://medium.com/nlplanet/two-minutes-nlp-learn-the-rouge-metric-by-examples-f179cc285499>
- <https://avinashselvam.medium.com/llm-evaluation-metrics-bleu-rouge-and-meteor-explained-a5d2b129e87f>
- <https://medium.com/explorations-in-language-and-learning/metrics-for-nlg-evaluation-c89b6a781054>
- <https://aclanthology.org/W04-1013/> (optional further reading)
- <https://aclanthology.org/2021.wmt-1.57/> (optional further reading)